

Term Project - Wine Quality

```
In [9]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from scipy import stats

from ucimlrepo import fetch_ucirepo

# fetch dataset
wine_quality = fetch_ucirepo(id=186)

# Combine 'features' and 'targets' DataFrames
complete_df = pd.concat([wine_quality['data']['features'], wine_quality['data']['ta

# Add 'color' column manually
num_red_wines = 1599 # Number of red wines in the dataset
complete_df['color'] = ['red'] * num_red_wines + ['white'] * (len(complete_df) - nu

# Separate the data based on wine color
red_wine = complete_df[complete_df['color'] == 'red']
white_wine = complete_df[complete_df['color'] == 'white']

# Calculate the mean quality for each color
mean_quality_red = red_wine['quality'].mean()
mean_quality_white = white_wine['quality'].mean()

print(f"Mean Quality for Red Wine: {mean_quality_red}")
print(f"Mean Quality for White Wine: {mean_quality_white}")

# Load and prepare the data
complete_df = pd.concat([wine_quality['data']['features'], wine_quality['data']['ta
num_red_wines = 1599
complete_df['color'] = ['red'] * num_red_wines + ['white'] * (len(complete_df) - nu

# Function to analyze a variable
def analyze_variable(variable_name):
    # Separate data by color
    red_wine = complete_df[complete_df['color'] == 'red']
    white_wine = complete_df[complete_df['color'] == 'white']

    # Plot histogram
    plt.figure(figsize=(10, 5))
    sns.histplot(red_wine[variable_name], color='red', label='Red Wine', kde=True,
    sns.histplot(white_wine[variable_name], color='green', label='White Wine', kde=
    plt.title(f'Histogram of {variable_name.capitalize()}')
    plt.xlabel(variable_name.capitalize())
    plt.ylabel('Density')
    plt.legend()
    plt.show()
```

```

# Calculate descriptive statistics
for color, wine_df in [('Red', red_wine), ('White', white_wine)]:
    mean_val = wine_df[variable_name].mean()
    median_val = wine_df[variable_name].median()
    mode_val = wine_df[variable_name].mode().tolist()
    std_val = wine_df[variable_name].std()
    variance_val = wine_df[variable_name].var()
    range_val = wine_df[variable_name].max() - wine_df[variable_name].min()
    skewness = wine_df[variable_name].skew()

# Identify outliers using IQR
Q1 = wine_df[variable_name].quantile(0.25)
Q3 = wine_df[variable_name].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = wine_df[(wine_df[variable_name] < lower_bound) | (wine_df[variab

print(f'\n{color} Wine - {variable_name.capitalize()} Statistics:')
print(f'Mean: {mean_val:.3f}')
print(f'Median: {median_val:.3f}')
print(f'Mode(s): {mode_val}')
print(f'Standard Deviation: {std_val:.3f}')
print(f'Variance: {variance_val:.3f}')
print(f'Range: {range_val:.3f}')
print(f'Skewness: {skewness:.3f}')
print(f'Number of Outliers: {len(outliers)}')
print(f'Outliers: {outliers.values}')

# Calculate Cohen's d for effect size between red and white wines
red_data = red_wine[variable_name]
white_data = white_wine[variable_name]
n1 = len(red_data)
n2 = len(white_data)
var1 = red_data.var()
var2 = white_data.var()

# Updated pooled standard deviation
pooled_std = np.sqrt((n1 * var1 + n2 * var2) / (n1 + n2))
cohen_d = (red_data.mean() - white_data.mean()) / pooled_std

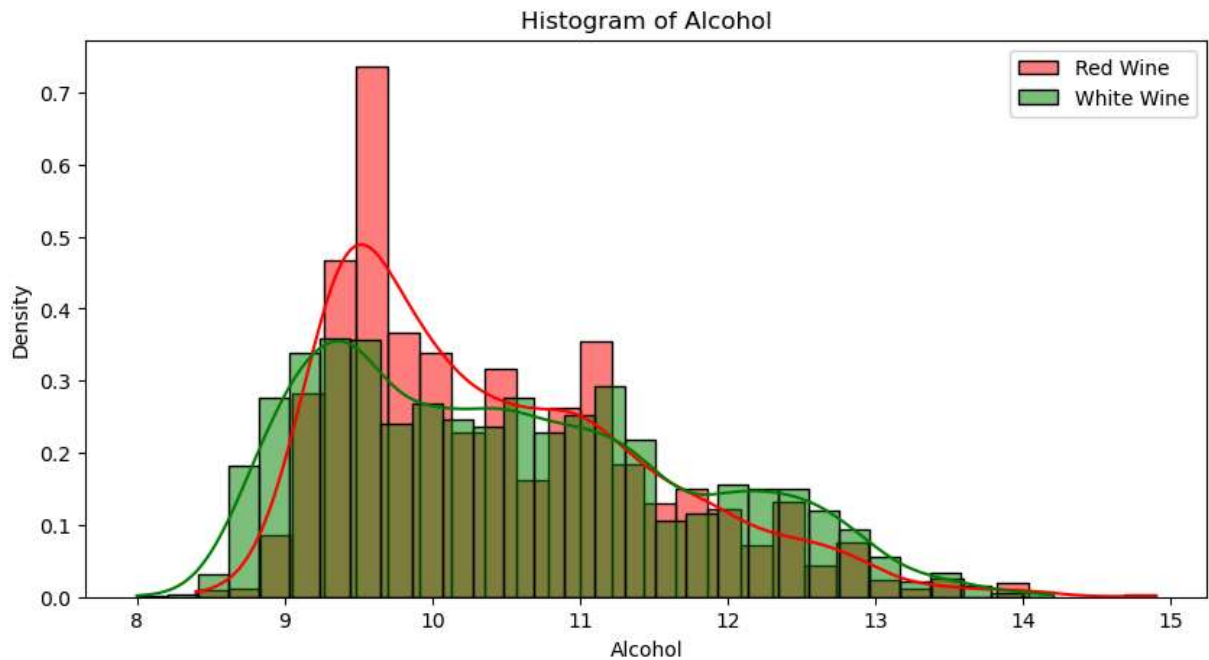
print(f"\nCohen's d for {variable_name.capitalize()} (Red vs. White): {cohen_d:

# Analyzing the 'alcohol' variable
analyze_variable('alcohol')

```

Mean Quality for Red Wine: 5.6360225140712945

Mean Quality for White Wine: 5.87790935075541



Red Wine - Alcohol Statistics:

Mean: 10.423

Median: 10.200

Mode(s): [9.5]

Standard Deviation: 1.066

Variance: 1.136

Range: 6.500

Skewness: 0.861

Number of Outliers: 13

Outliers: [14. 14. 14. 14. 14.9 14.
 13.6 13.6 13.6 14. 14. 13.56666667
 13.6]

White Wine - Alcohol Statistics:

Mean: 10.514

Median: 10.400

Mode(s): [9.4]

Standard Deviation: 1.231

Variance: 1.514

Range: 6.200

Skewness: 0.487

Number of Outliers: 0

Outliers: []

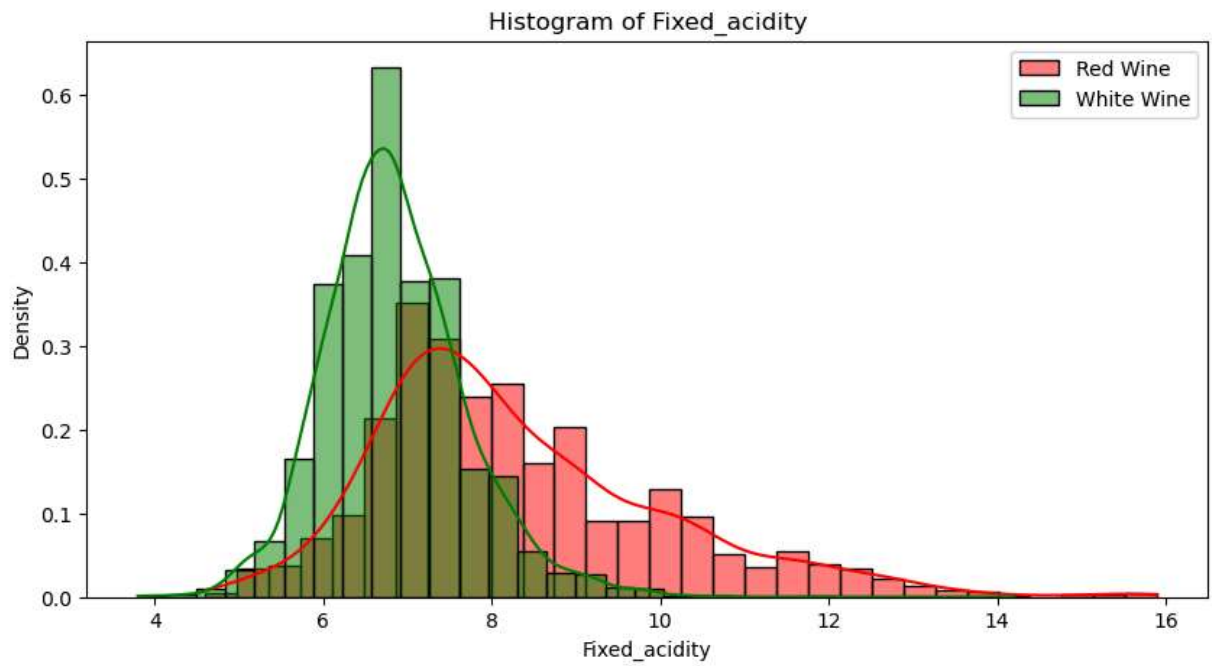
Cohen's d for Alcohol (Red vs. White): -0.077

A Cohen's d of -0.077 is very close to 0 and shows a very small effect size between red and white wines in terms of their alcohol content. The negative sign tells us the direction of the difference. So on average, the alcohol content of red wine is slightly lower than white wine.

Outliers: Alcohol content in wines generally are between 8-15%, while some varieties could be higher. If any values are significantly over 15%, they could be indicating a potential error, or a unique type of wine. Anything below 8% are unusual but not necessarily errors. To this end, I would keep values within 8-15% but remove values over 15%.

```
In [32]: # Analyze other variables in the same way:
```

```
# Analyzing the 'fixed acidity' variable  
analyze_variable('fixed_acidity')
```



Red Wine - Fixed_acidity Statistics:

Mean: 8.320

Median: 7.900

Mode(s): [7.2]

Standard Deviation: 1.741

Variance: 3.031

Range: 11.300

Skewness: 0.983

Number of Outliers: 49

Outliers: [12.8 12.8 15. 15. 12.5 13.3 13.4 12.4 12.5 13.8 13.5 12.6 12.5 12.8
12.8 14. 13.7 13.7 12.7 12.5 12.8 12.6 15.6 12.5 13. 12.5 13.3 12.4
12.5 12.9 14.3 12.4 15.5 15.5 15.6 13. 12.7 13. 12.7 12.4 12.7 13.2
13.2 13.2 15.9 13.3 12.9 12.6 12.6]

White Wine - Fixed_acidity Statistics:

Mean: 6.855

Median: 6.800

Mode(s): [6.8]

Standard Deviation: 0.844

Variance: 0.712

Range: 10.400

Skewness: 0.648

Number of Outliers: 119

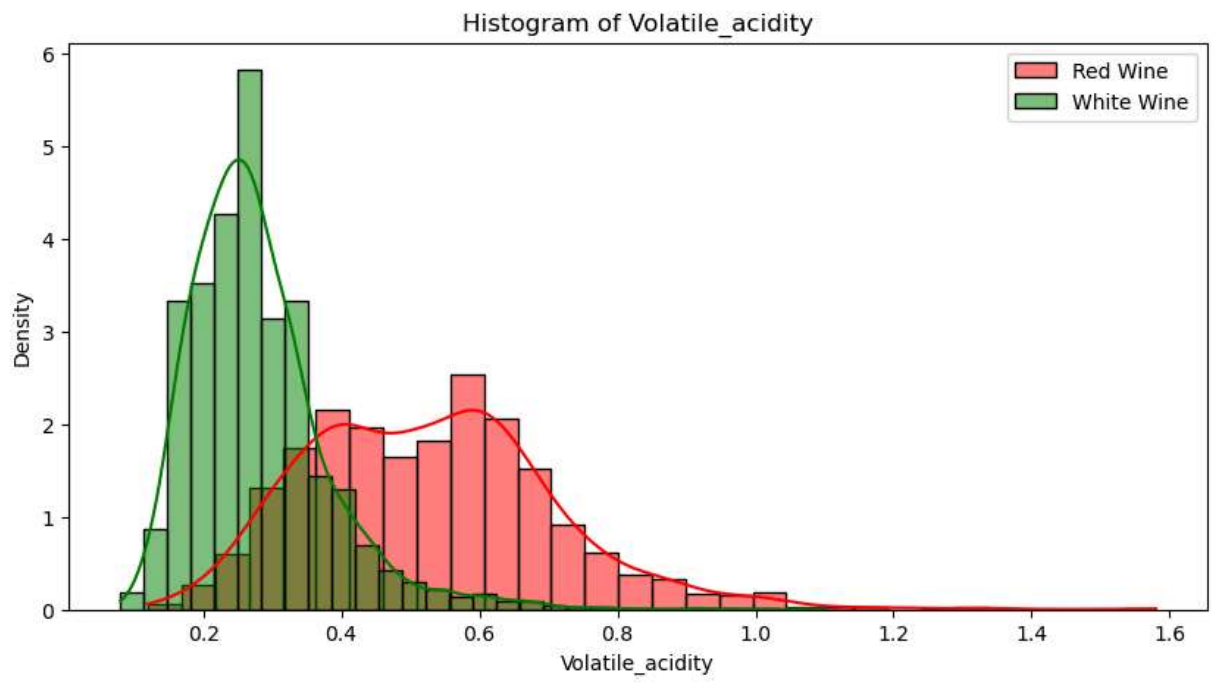
Outliers: [9.8 9.8 10.2 9.1 10. 9.2 9.2 9. 9.1 9.2 10.3 9.4 9.2 9.8
9.6 9.2 9. 9.3 9.2 9.1 8.9 9.8 8.9 9.2 9.7 9.4 10.3 9.6
9. 9.7 9.2 9.4 9.6 9.2 9. 9.2 10.7 10.7 9. 9.2 9.8 9.2
14.2 8.9 8.9 9.1 9.1 9.8 9. 9.3 8.9 9. 9. 8.9 9. 9.3
9.2 9.6 9.4 9.4 10. 8.9 8.9 10. 9.2 9.2 9.2 9.9 9.5 9.
9. 8.9 9.5 11.8 9.4 9.1 9.8 9.9 9.2 8.9 9.2 9.4 9.4 9.4
4.6 8.9 9.4 9.2 9.2 9.8 9. 9. 9. 8.9 8.9 4.5 9.2 9.6
4.2 9.7 9.7 9. 4.2 9.4 8.9 8.9 8.9 4.7 4.7 3.8 4.4 4.7
9. 9. 4.7 4.4 3.9 4.7 4.4]

Cohen's d for Fixed_acidity (Red vs. White): 1.293

High outliers of fixed acidity for both red and white wines could represent certain wine-making techniques or grape varieties that lead to unusually high acidity. However, extremely low values (like 3.8) are uncommon and may be considered data entry errors and should be examined closer.

Cohen's d of 1.293 indicates a significant difference in fixed acidity between red and white wines, supporting the idea that red wines have higher acidity levels than white wines.

```
In [34]: # Analyzing the 'volatile acidity' variable  
analyze_variable('volatile_acidity')
```



Red Wine - Volatile_acidity Statistics:

Mean: 0.528

Median: 0.520

Mode(s): [0.6]

Standard Deviation: 0.179

Variance: 0.032

Range: 1.460

Skewness: 0.672

Number of Outliers: 19

Outliers: [1.13 1.02 1.07 1.33 1.33 1.04 1.09 1.04 1.24 1.185 1.02 1.035
1.025 1.115 1.02 1.02 1.58 1.18 1.04]

White Wine - Volatile_acidity Statistics:

Mean: 0.278

Median: 0.260

Mode(s): [0.28]

Standard Deviation: 0.101

Variance: 0.010

Range: 1.020

Skewness: 1.577

Number of Outliers: 186

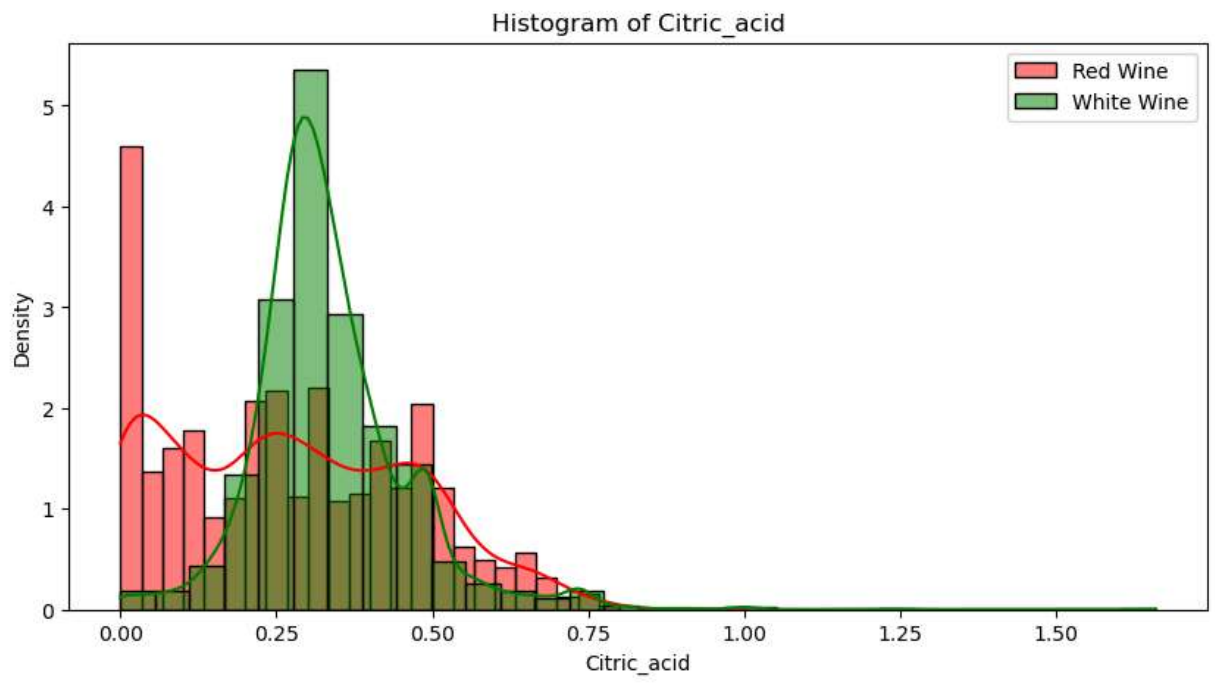
Outliers: [0.66 0.66 0.67 0.54 0.595 0.67 0.53 0.54 0.57 0.685 0.495 0.64
0.52 0.58 0.585 0.59 0.6 0.58 0.59 0.55 0.905 0.55 0.49 0.55
0.52 0.6 0.55 0.51 0.62 0.51 0.56 0.57 0.67 0.5 0.56 0.56
0.655 0.595 0.705 0.52 0.55 0.6 0.64 0.68 0.49 0.51 0.55 0.52
0.5 0.55 0.6 0.61 0.61 0.61 0.66 0.57 0.5 0.5 0.59 0.58
0.54 0.58 0.57 0.64 0.56 0.49 0.49 0.67 0.55 0.56 0.52 0.52
0.85 0.51 0.62 0.51 0.53 0.64 0.55 0.49 0.49 0.61 0.545 0.62
0.49 0.5 0.49 0.49 0.55 0.49 0.91 0.53 0.49 0.71 1.005 0.49
0.55 0.55 0.76 0.5 0.93 0.49 0.495 0.695 0.705 0.815 0.56 0.56
0.56 0.51 0.54 0.54 0.5 0.615 0.5 0.52 0.6 0.68 0.655 0.51
0.51 0.615 0.615 0.965 0.74 0.53 0.78 0.68 0.64 0.54 0.75 0.64
0.64 0.655 0.58 0.52 0.53 0.6 0.53 0.58 0.67 0.61 0.73 0.65
0.58 1.1 0.5 0.5 0.5 0.65 0.52 0.55 0.585 0.56 0.555 0.555
0.54 0.61 0.55 0.53 0.66 0.615 0.5 0.62 0.5 0.49 0.51 0.51
0.54 0.61 0.695 0.695 0.63 0.63 0.69 0.69 0.59 0.62 0.785 0.76
0.5 0.54 0.52 0.6 0.54 0.53]

Cohen's d for Volatile_acidity (Red vs. White): 2.001

When it comes to volatile acidity, we should consider removing values higher than 1.2 for red wines and 1.1 for white wines since they are unrealistic based on typical wine characteristics. However, moderate outliers (1 for red wines and 0.7 for white wines) should be retained since they could offer insights into variations in quality or taste.

Similar to fixed acidity, the effect size of volatile acidity is very large, suggesting that red wines have substantially higher volatile acidity compared to white wines.

```
In [36]: # Analyzing the 'citric acid' variable  
analyze_variable('citric_acid')
```



Red Wine - Citric_acid Statistics:

Mean: 0.271

Median: 0.260

Mode(s): [0.0]

Standard Deviation: 0.195

Variance: 0.038

Range: 1.000

Skewness: 0.318

Number of Outliers: 1

Outliers: [1.]

White Wine - Citric_acid Statistics:

Mean: 0.334

Median: 0.320

Mode(s): [0.3]

Standard Deviation: 0.121

Variance: 0.015

Range: 1.660

Skewness: 1.282

Number of Outliers: 270

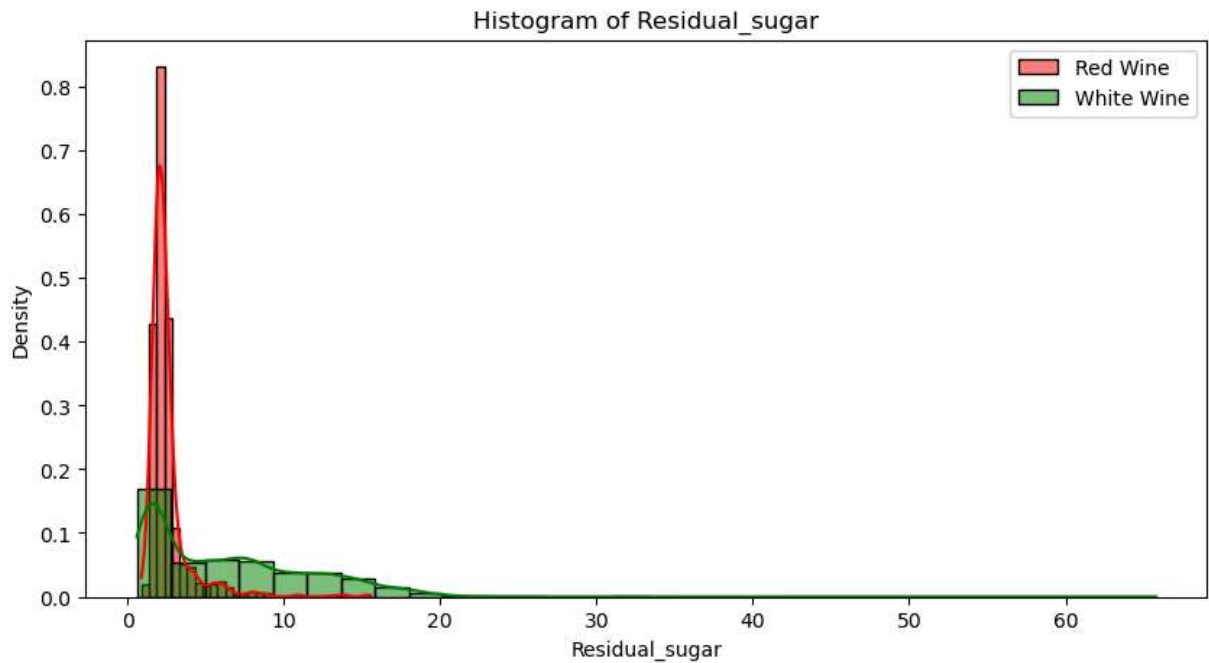
Outliers: [0.62 0.04 0.59 0.07 0.03 0.61 0.62 0.63 0.61 0.62 0.63 0.66 0.66 0.
0.04 0.67 0.67 0.04 0.04 0.07 0.88 0.08 0.59 0.07 0.07 0.07 0.07 0.58
0.7 0. 0. 0.6 0.07 0.09 0.04 0.62 0.58 0.62 0.7 0.62 0.62 0.58
0.02 0.65 0.65 0.71 0.66 0.66 0.07 0.06 0.07 0.06 0.68 0.68 0.68 0.68
0.06 0.72 0.69 0.58 0.7 1.66 0.04 0.63 0.6 0. 0.08 0.58 0.58 0.05
0.58 0. 0. 0.65 0.58 0. 0.05 0.05 0.62 0.62 0.58 0.58 1. 0.09
0.01 0.71 0.71 0.6 0.06 0.74 0.81 0.69 0.58 0.69 0. 0.07 0.64 0.72
0.73 0.65 0.68 0.65 0.74 0.71 0.59 0.68 0.08 0.72 0.64 0.02 0.74 0.74
0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74
0.74 0.74 0.74 0.74 0.74 0.99 0.74 0.74 0.74 0.74 0.74 0.74 0.74 0.74
0.74 0.74 0.74 0.74 0.74 0.74 0.01 0.74 0.01 0.74 0.74 1. 0.04 0.58
0.07 1. 0. 0.58 0.61 0.61 0.61 0.02 0.67 0.67 0.67 0.58 0.65 0.58
0.09 0.08 0.71 0.04 0.03 0.05 0.64 0.64 0.58 0.58 0.81 0.58 0.61 0.62
0.59 0. 0.04 0.63 0.73 0.68 0.09 0.78 0.79 0.09 0.64 0.65 0.65 0.
0.73 0.73 0.64 0.6 0.71 0.72 0.82 0.07 0.58 0.58 1. 0.66 0.8 0.8
1.23 0.59 0.02 0. 1. 0.62 0. 0.71 0.71 0.71 0.61 0.61 0. 0.6
0.58 0.09 0.09 0.72 0.62 0.62 0.79 0.82 0.67 0.01 0.01 0.86 0.61 0.02
0.05 0. 0.69 0.69 0.59 0.01 0.66 0.66 0.78 0. 0.04 0.91 0.91 0.06
0.06 0.04 0.04 0.74 0.09 0.09 0.6 0.62 0.73 0. 0.09 0. 0.09 0.67
0.01 0.09 0. 0.02]

Cohen's d for Citric_acid (Red vs. White): -0.443

Red wine has just 1 outlier with a value of 1, while white wine has 270 outliers with values both high and low. The outlier of 1 is high for red wine, where citric acid levels tend to be lower. However, it might represent a genuine extreme due to the grape variety or fermentation process. So in this case, we will retain this outlier. For the white wines, extremely high values over 1.66 might be a measurement error, while the outliers around 0.7 to 1 may just be true extremes.

Cohen's d for citric acid is -0.443 tells us that red wines have slightly lower citric acid levels.

```
In [38]: # Analyzing the 'residual_sugar' variable
analyze_variable('residual_sugar')
```



Red Wine - Residual_sugar Statistics:

Mean: 2.539

Median: 2.200

Mode(s): [2.0]

Standard Deviation: 1.410

Variance: 1.988

Range: 14.600

Skewness: 4.541

Number of Outliers: 155

Outliers: [6.1 6.1 3.8 3.9 4.4 10.7 5.5 5.9 5.9 3.8 5.1 4.65
 4.65 5.5 5.5 5.5 5.5 7.3 7.2 3.8 5.6 4. 4. 4.
 4. 7. 4. 4. 6.4 5.6 5.6 11. 11. 4.5 4.8 5.8
 5.8 3.8 4.4 6.2 4.2 7.9 7.9 3.7 4.5 6.7 6.6 3.7
 5.2 15.5 4.1 8.3 6.55 6.55 4.6 6.1 4.3 5.8 5.15 6.3
 4.2 4.2 4.6 4.2 4.6 4.3 4.3 7.9 4.6 5.1 5.6 5.6
 6. 8.6 7.5 4.4 4.25 6. 3.9 4.2 4. 4. 4. 6.6
 6. 6. 3.8 9. 4.6 8.8 8.8 5. 3.8 4.1 5.9 4.1
 6.2 8.9 4. 3.9 4. 8.1 8.1 6.4 6.4 8.3 8.3 4.7
 5.5 5.5 4.3 5.5 3.7 6.2 5.6 7.8 4.6 5.8 4.1 12.9
 4.3 13.4 4.8 6.3 4.5 4.5 4.3 4.3 3.9 3.8 5.4 3.8
 6.1 3.9 5.1 5.1 3.9 15.4 15.4 4.8 5.2 5.2 3.75 13.8
 13.8 5.7 4.3 4.1 4.1 4.4 3.7 6.7 13.9 5.1 7.8]

White Wine - Residual_sugar Statistics:

Mean: 6.391

Median: 5.200

Mode(s): [1.2]

Standard Deviation: 5.072

Variance: 25.726

Range: 65.200

Skewness: 1.077

Number of Outliers: 7

Outliers: [23.5 31.6 31.6 65.8 26.05 26.05 22.6]

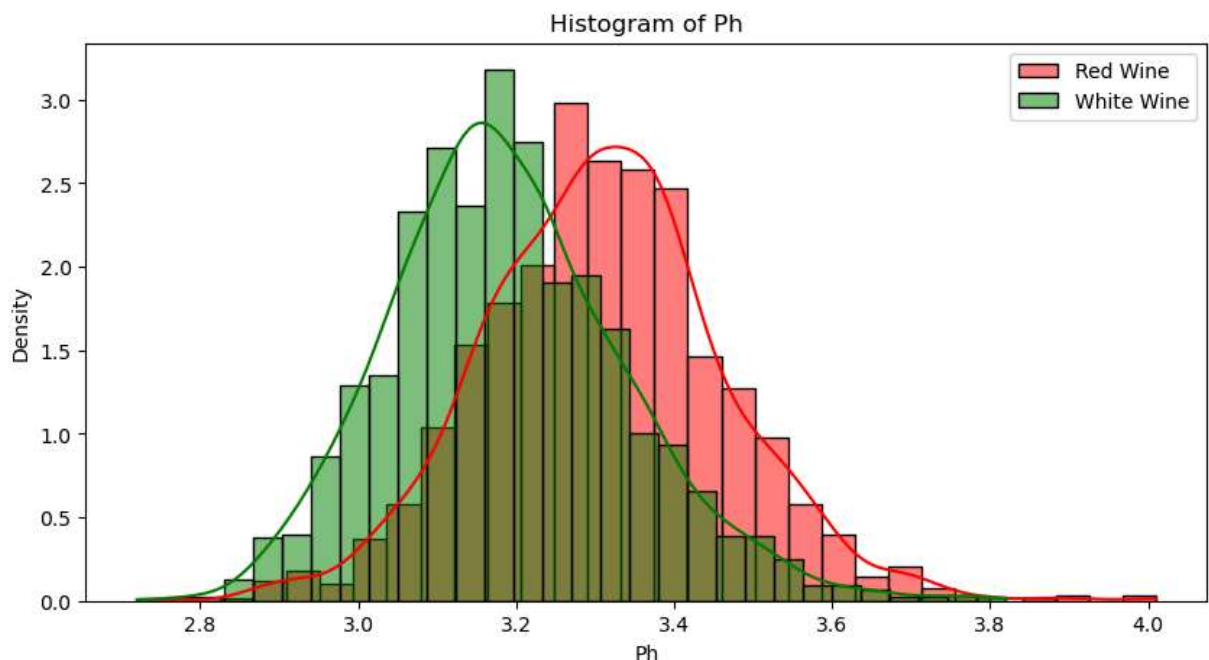
Cohen's d for Residual_sugar (Red vs. White): -0.864

For red wines, there are 155 outliers, which is not surprising since red wines tend to have lower residual sugar. Considering this, outliers between 4 and 8 may be considered true extremes that indicate genuine variations due to fermentation methods or sugar additions during the process. However, values beyond 15 are well above the average residual sugar for red wines and could be errors.

When it comes to white wines, the outliers around 20-30 could actually be wines with added sugar or naturally sweet wines (think dessert wines), so they should be retained as meaningful data points. However, we should consider removing the 65.8 value which is more unrealistic and could be a measurement error.

The cohen's d of -0.864 tells us there is a moderate difference in residual sugar between red and white wines. White wines tend to have higher sugar levels.

```
In [40]: # Analyzing the 'pH' variable  
analyze_variable('pH')
```



Red Wine - Ph Statistics:

Mean: 3.311

Median: 3.310

Mode(s): [3.3]

Standard Deviation: 0.154

Variance: 0.024

Range: 1.270

Skewness: 0.194

Number of Outliers: 35

Outliers: [3.9 3.75 3.85 2.74 3.69 3.69 2.88 2.86 3.74 2.92 2.92 2.92 3.72 2.87
2.89 2.89 2.92 3.9 3.71 3.69 3.69 3.71 3.71 2.89 2.89 3.78 3.7 3.78
4.01 2.9 4.01 3.71 2.88 3.72 3.72]

White Wine - Ph Statistics:

Mean: 3.188

Median: 3.180

Mode(s): [3.14]

Standard Deviation: 0.151

Variance: 0.023

Range: 1.100

Skewness: 0.458

Number of Outliers: 75

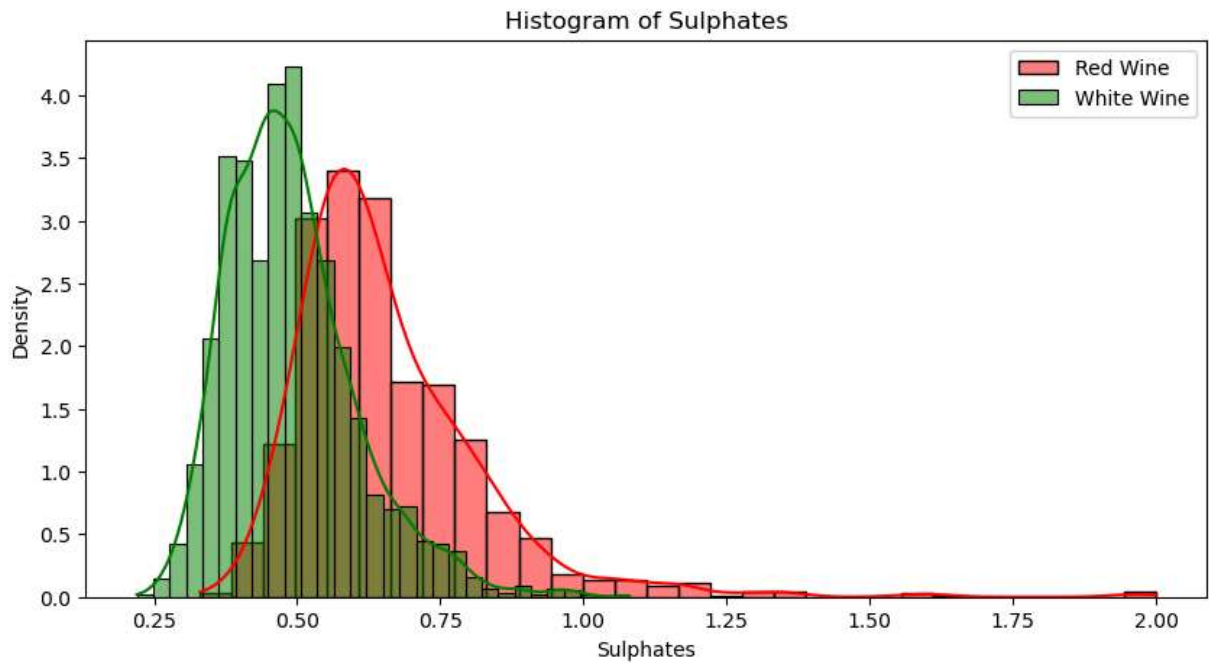
Outliers: [3.69 3.63 3.72 3.61 3.64 3.64 3.72 3.72 3.58 3.58 3.66 3.59 2.74 3.82
3.81 3.65 3.65 3.59 3.77 3.62 3.63 3.58 3.58 3.65 3.74 2.8 3.6 3.6
2.72 3.6 2.79 2.79 3.57 3.8 3.6 3.6 3.68 3.63 3.63 2.77 3.63 3.6
3.6 3.61 3.61 3.59 3.79 3.59 3.68 3.59 3.66 3.7 3.74 3.8 3.57 3.57
3.57 3.65 3.58 2.8 3.77 3.76 3.69 3.66 3.59 2.79 3.75 3.63 3.75 3.76
3.66 3.66 2.8 3.67 3.57]

Cohen's d for Ph (Red vs. White): 0.809

When it comes to pH, we should consider removing extremely high and low values, such as outliers below 2.8 and outliers above 3.8. However, values that are more moderate should be retained for analysis since they could represent a natural acidity difference that impacts flavor and stability.

Based on Cohen's d, we know that red wines generally have higher pH levels.

```
In [158... # Analyzing the 'sulphates' variable  
analyze_variable('sulphates')
```



Red Wine - Sulphates Statistics:

Mean: 0.658

Median: 0.620

Mode(s): [0.6]

Standard Deviation: 0.170

Variance: 0.029

Range: 1.670

Skewness: 2.429

Number of Outliers: 59

Outliers: [1.56 1.28 1.08 1.2 1.12 1.28 1.14 1.95 1.22 1.95 1.98 1.31 2. 1.08
1.59 1.02 1.03 1.61 1.09 1.26 1.08 1. 1.36 1.18 1.13 1.04 1.11 1.13
1.07 1.06 1.06 1.05 1.06 1.04 1.05 1.02 1.14 1.02 1.36 1.36 1.05 1.17
1.62 1.06 1.18 1.07 1.34 1.16 1.1 1.15 1.17 1.17 1.33 1.18 1.17 1.03
1.17 1.1 1.01]

White Wine - Sulphates Statistics:

Mean: 0.490

Median: 0.470

Mode(s): [0.5]

Standard Deviation: 0.114

Variance: 0.013

Range: 0.860

Skewness: 0.977

Number of Outliers: 124

Outliers: [0.77 0.84 0.77 0.79 0.85 0.78 0.79 0.79 0.79 0.77 0.78 0.85 0.96 0.97
0.82 0.82 0.77 0.95 0.95 0.77 0.95 0.82 0.82 0.9 0.88 0.88 0.79 0.8
0.8 0.78 0.78 0.87 0.86 0.9 0.9 0.78 0.79 0.81 0.81 0.77 0.82 0.79
0.79 0.77 0.82 0.92 0.79 0.79 0.82 0.82 0.82 0.82 0.82 0.79 0.78 0.79
0.77 0.77 0.77 0.98 1.06 0.88 0.88 0.88 0.8 0.78 1. 0.8 0.9 0.9
0.89 0.94 0.99 0.86 0.84 0.95 0.84 0.84 0.81 0.8 0.87 0.82 0.78 0.78
0.78 0.78 0.78 0.77 0.85 0.78 0.78 0.88 0.88 0.78 0.78 0.78 0.78 0.79
0.77 0.77 0.83 0.83 0.81 0.81 0.98 0.98 0.98 0.98 0.79 0.79 0.78 0.82
0.98 0.77 0.96 1.01 0.77 0.96 0.77 0.92 0.94 0.95 1.08 0.79]

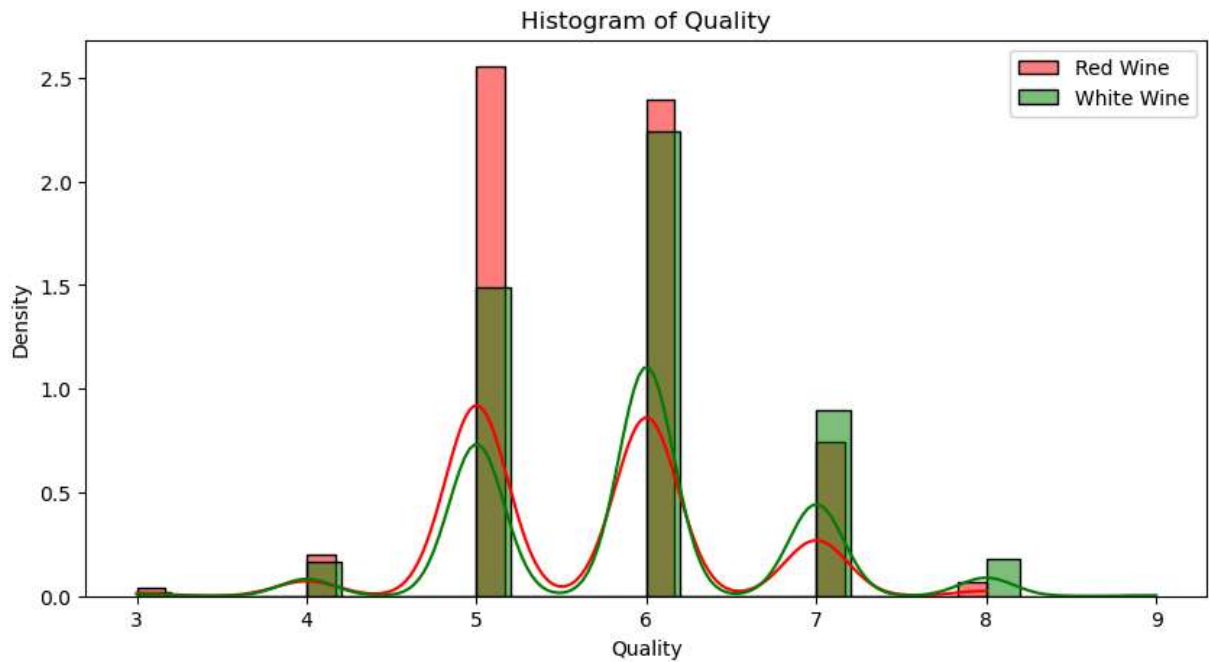
Cohen's d for Sulphates (Red vs. White): 1.295

Sulphates are used as additives in wine and has various benefits, including preventing oxidation and bacterial growth, promoting the fermentation process and releasing desirable compounds from the grapes.

Based on the histogram and cohen's d, we know that red wine tends to have a higher level of sulphates compared to white wine.

In [167...

```
# Analyzing the 'quality' variable  
analyze_variable('quality')
```



Red Wine - Quality Statistics:

Mean: 5.636

Median: 6.000

Mode(s): [5]

Standard Deviation: 0.808

Variance: 0.652

Range: 5.000

Skewness: 0.218

Number of Outliers: 28

Outliers: [8 8 8 8 8 3 8 8 8 3 8 3 3 8 8 8 8 8 3 3 8 8 3 3 3 8]

White Wine - Quality Statistics:

Mean: 5.878

Median: 6.000

Mode(s): [6]

Standard Deviation: 0.886

Variance: 0.784

Range: 6.000

Skewness: 0.156

Number of Outliers: 200

Outliers: [8 8 8 8 8 8 8 3 3 8 8 8 3 8 8 8 8 8 3 8 8 8 8 8 3 9 8 8 8 9 9 8 8 8 8 8
8 8 8 8 3 9 8 8 8 8 8 3 8 8 8 8 8 8 8 8 8 3 8 8 8 8 8 8 8 8 8 8 8 8 8 8
3 8 3 8 8 8 9 8 8 8 3 8 8 8 8 8 3 8 8 8 8 8 3 8 8 8 3 8 8 8 8 8 8 8 8
8 3 8 8 8 8 8 8 8 3 8 8
8 3 8 8 8 3 8 3 8 8 8 8 8 8
8 8 8 8 8 8 8 8 8 3 8 8 8 8 8]

Cohen's d for Quality (Red vs. White): -0.279

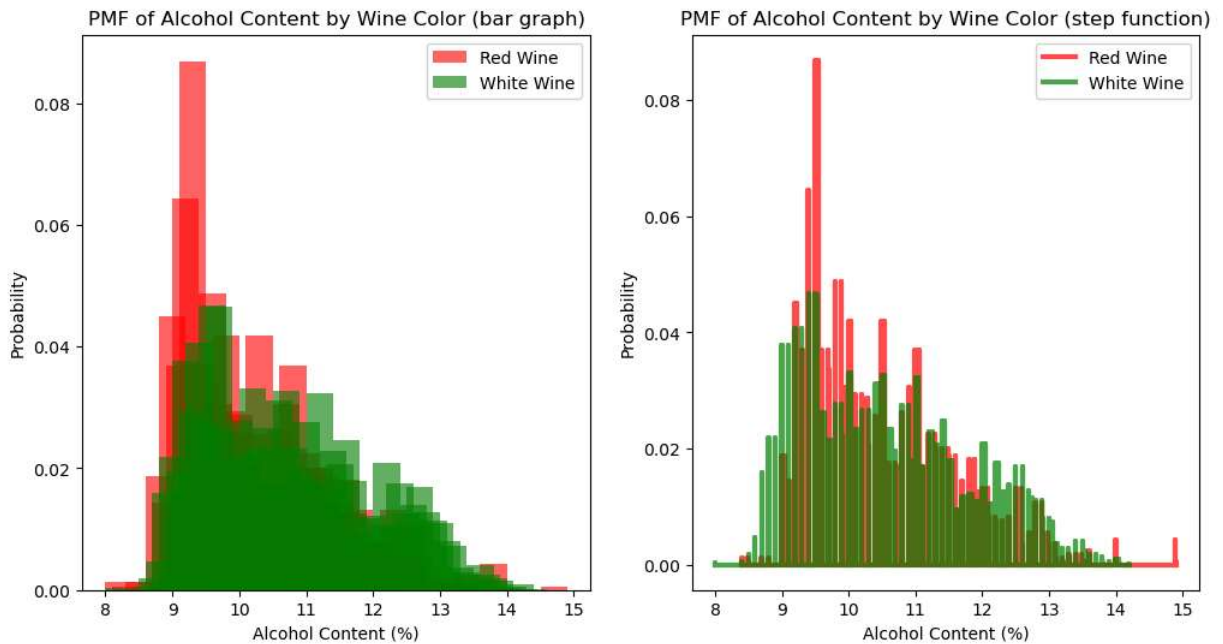
```
In [66]: import thinkstats2
import thinkplot

# Separate the alcohol content for red and white wines
red_wine_alcohol = complete_df[complete_df['color'] == 'red']['alcohol']
white_wine_alcohol = complete_df[complete_df['color'] == 'white']['alcohol']

# Create PMFs
red_pmf = thinkstats2.Pmf(red_wine_alcohol, label='Red Wine')
white_pmf = thinkstats2.Pmf(white_wine_alcohol, label='White Wine')

# Plot the PMFs as bar graphs
thinkplot.PrePlot(2, cols=2)
thinkplot.Hist(red_pmf, align='right', width=0.4, color='red')
thinkplot.Hist(white_pmf, align='left', width=0.4, color='green')
thinkplot.Config(xlabel='Alcohol Content (%)', ylabel='Probability', title='PMF of

thinkplot.PrePlot(2)
thinkplot.SubPlot(2)
thinkplot.Pmf(red_pmf, color='red', label='Red Wine')
thinkplot.Pmf(white_pmf, color='green', label='White Wine')
thinkplot.Show(xlabel='Alcohol Content (%)', ylabel='Probability', title='PMF of Al
```



<Figure size 800x600 with 0 Axes>

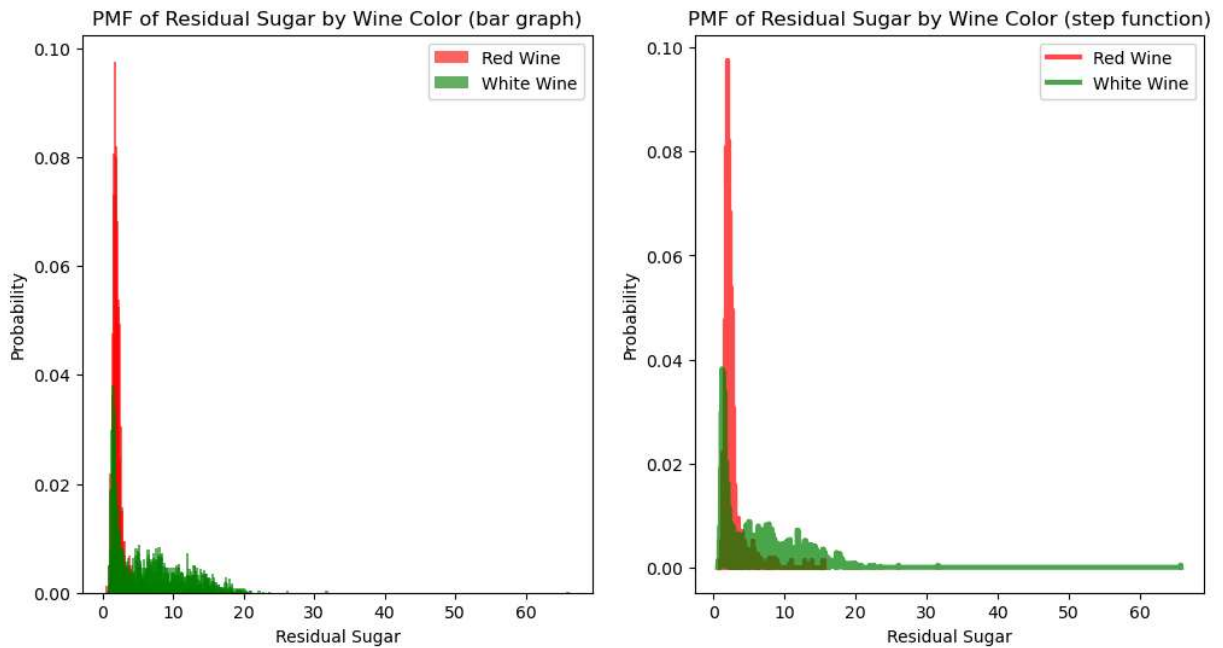
The PMF indicates that red wines tend to have a lower alcohol content, with a peak between 9.5 to 10%. On the other hand, white wines have a broader distribution of alcohol content, with a peak around 10-11% and a more even spread up until about 13%. This suggests that winemaking practices for whites may involve more flexibility in alcohol levels. There is significant overlap in alcohol content between red and white wines, but white wines show more occurrences of higher alcohol content from this dataset. When looking at the tails of the distribution, we can see higher alcohol percentages are more common in white wines, and red wines taper off earlier.

```
In [72]: # Separate the alcohol content for red and white wines
red_wine_sugar = complete_df[complete_df['color'] == 'red']['residual_sugar']
white_wine_sugar = complete_df[complete_df['color'] == 'white']['residual_sugar']

# Create PMFs
red_pmf = thinkstats2.Pmf(red_wine_sugar, label='Red Wine')
white_pmf = thinkstats2.Pmf(white_wine_sugar, label='White Wine')

# Plot the PMFs as bar graphs
thinkplot.PrePlot(2, cols=2)
thinkplot.Hist(red_pmf, align='right', width=0.4, color='red')
thinkplot.Hist(white_pmf, align='left', width=0.4, color='green')
thinkplot.Config(xlabel='Residual Sugar', ylabel='Probability', title='PMF of Resid

thinkplot.PrePlot(2)
thinkplot.SubPlot(2)
thinkplot.Pmf(red_pmf, color='red', label='Red Wine')
thinkplot.Pmf(white_pmf, color='green', label='White Wine')
thinkplot.Show(xlabel='Residual Sugar', ylabel='Probability', title='PMF of Residua
```

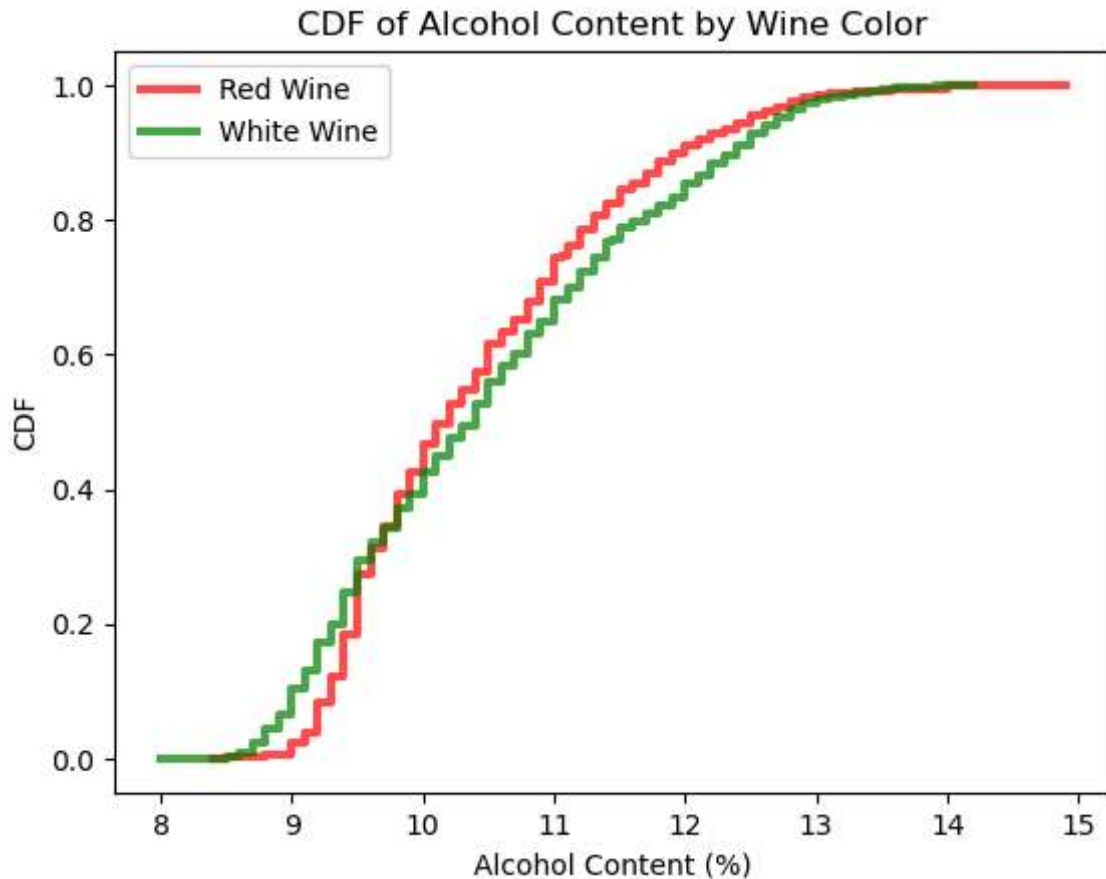


<Figure size 800x600 with 0 Axes>

The long tail that can be seen for the white wines tell us that white wines have much higher residual sugar levels compared to red wines. Red wines rarely exceed 2 grams per liter of residual sugar, signifying a pattern of drier red wines. This difference is consistent with common wine knowledge, where white wines are often available in both dry and sweet, while red wines are usually made in drier varieties.

```
In [69]: # Create CDFs
red_cdf = thinkstats2.Cdf(red_wine_alcohol, label='Red Wine')
white_cdf = thinkstats2.Cdf(white_wine_alcohol, label='White Wine')

# Plot the CDFs
thinkplot.PrePlot(2)
thinkplot.Cdf(red_cdf, color='red')
thinkplot.Cdf(white_cdf, color='green')
thinkplot.Config(xlabel='Alcohol Content (%)', ylabel='CDF', title='CDF of Alcohol')
thinkplot.Show()
```

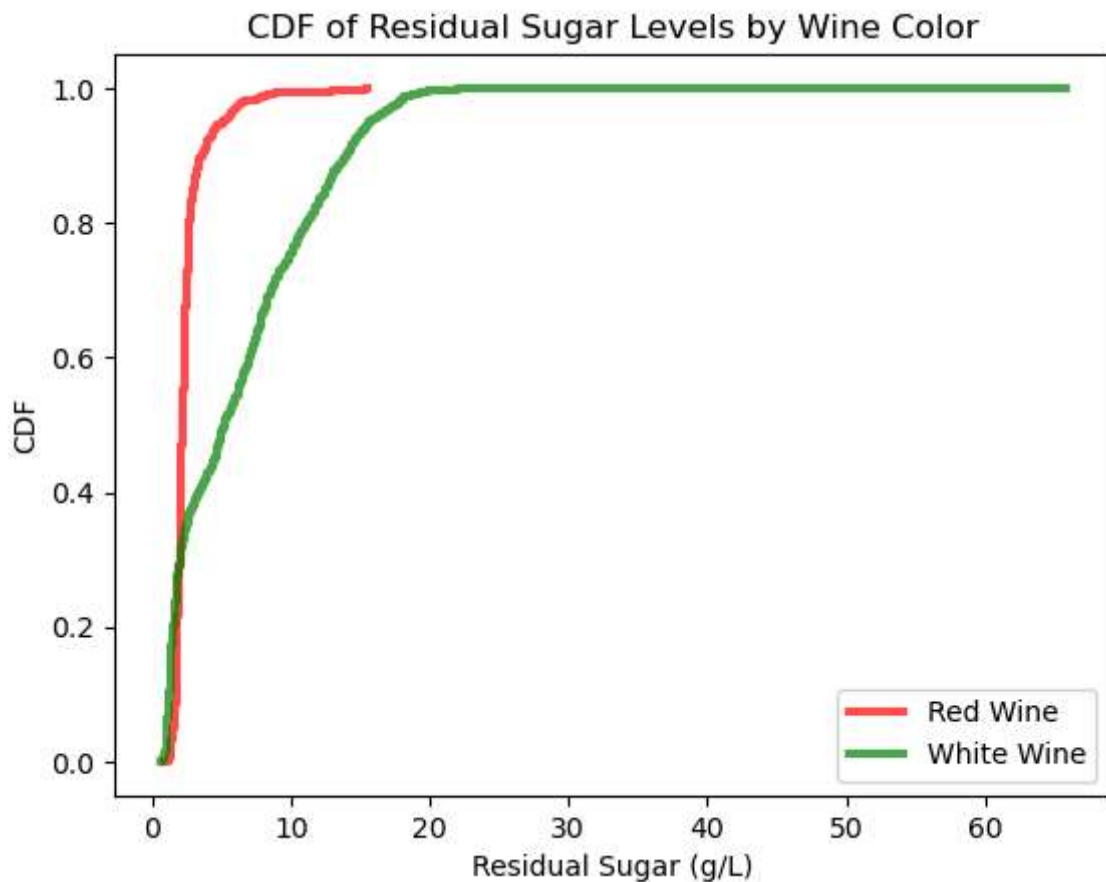


<Figure size 800x600 with 0 Axes>

The CDF for white wine is shifted to the right of the CDF for red wines, specifically around the 9.5-12% range, suggesting that in general white wines tend to have slightly higher alcohol content compared to red wines. The median for red wine is around 10% and 10.5% for white, which also reinforces the first point. The CDF for red wine rises more quickly in the lower range, showing that more red wines are clustered around lower alcohol levels. The CDF for white wines have a more gradual increase, showing a broader distribution of alcohol content.

```
In [78]: # Create CDFs
red_cdf = thinkstats2.Cdf(red_wine_sugar, label='Red Wine')
white_cdf = thinkstats2.Cdf(white_wine_sugar, label='White Wine')

# Plot the CDFs
thinkplot.PrePlot(2)
thinkplot.Cdf(red_cdf, color='red')
thinkplot.Cdf(white_cdf, color='green')
thinkplot.Config(xlabel='Residual Sugar (g/L)', ylabel='CDF', title='CDF of Residual Sugar by Wine Color')
thinkplot.Show()
```



<Figure size 800x600 with 0 Axes>

The CDF for red wine increases rapidly and reaches 1.0 at around 5g per liter of residual sugar, showing that most red wines in the dataset have low residual sugar levels. The CDF for white wines rise a lot more gradually, with some wines having values as high as 60g/L. Residual sugar is a distinguishing factor between red and white wines, with whites having a higher potential for sweetness and much greater variability.

This normal probability plot shows us that white and red wines both have linear characteristics but it doesn't quite fit a normal distribution, specifically when looking at the tails. This is in line with the findings we had earlier where we saw a slight skew, implying that there may be a different model (lognormal?) that could be a better fit.

In [115...

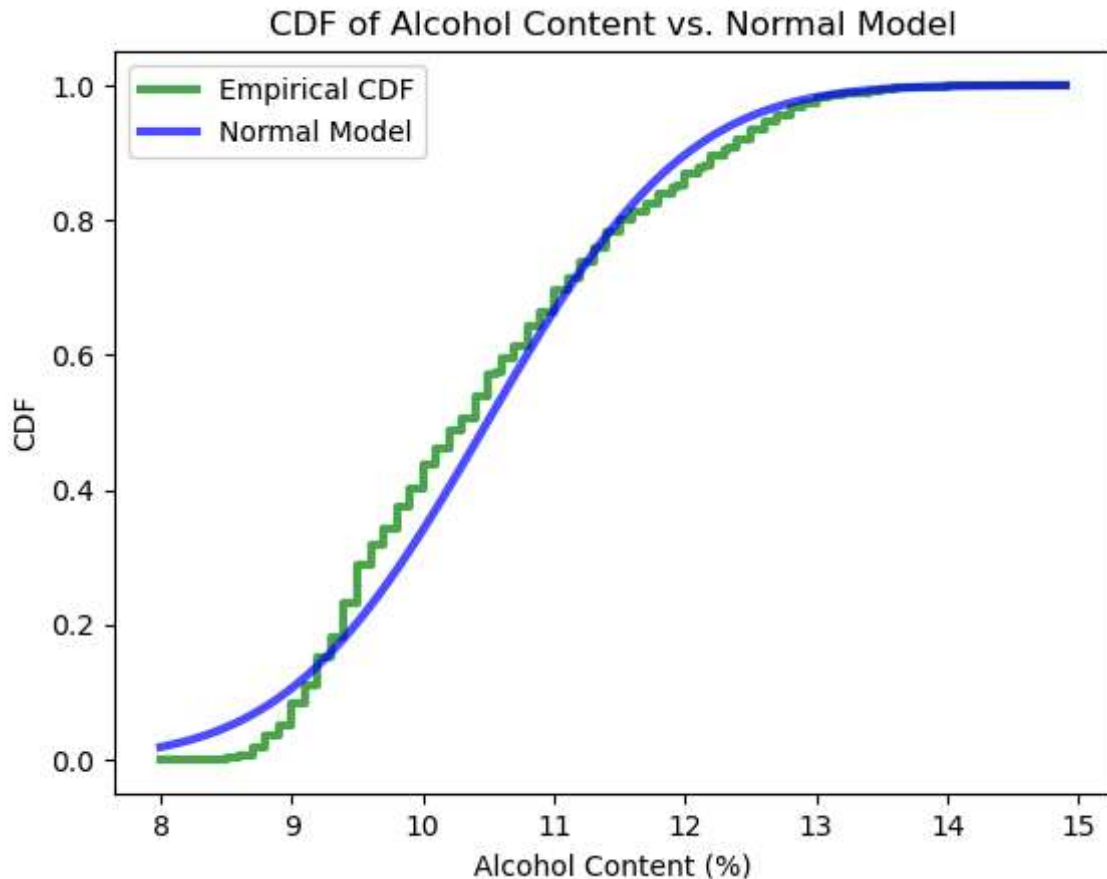
```
# Data: Alcohol content for both wines
alcohol_content = complete_df['alcohol']

# Calculate mean and standard deviation
mean = alcohol_content.mean()
std_dev = alcohol_content.std()

# Generate empirical CDF for alcohol content
ecdf = thinkstats2.Cdf(alcohol_content, label='Empirical CDF')

# Generate values for the normal CDF based on the mean and standard deviation
x = np.linspace(alcohol_content.min(), alcohol_content.max(), 100)
normal_cdf = stats.norm.cdf(x, mean, std_dev)
```

```
# Plot the empirical CDF and the normal CDF
thinkplot.Cdf(ecdf, label='Empirical CDF', color='green')
thinkplot.Plot(x, normal_cdf, label='Normal Model', color='blue')
thinkplot.Config(xlabel='Alcohol Content (%)', ylabel='CDF',
                  title='CDF of Alcohol Content vs. Normal Model')
thinkplot.Show()
```



<Figure size 800x600 with 0 Axes>

The empirical CDF deviates from the model, specifically at both ends. The normal model overestimates the CDF for the lower alcohol content and underestimates it for the higher alcohol content. Overall, this normal model provides a rough approximation but doesn't capture the actual distribution perfectly.

In [120...

```
# Data: Define variables for alcohol vs quality and residual sugar vs quality
alcohol = complete_df['alcohol']
quality = complete_df['quality']
residual_sugar = complete_df['residual_sugar']

# Scatter Plot 1: Alcohol Content vs. Quality
thinkplot.Scatter(alcohol, quality, alpha=0.5)
thinkplot.Config(xlabel="Alcohol Content (%)", ylabel="Quality",
                  title="Scatter Plot of Alcohol Content vs. Quality")
thinkplot.Show()

# Scatter Plot 2: Residual Sugar vs. Quality
thinkplot.Scatter(residual_sugar, quality, alpha=0.5)
```

```

thinkplot.Config(xlabel="Residual Sugar (g/L)", ylabel="Quality",
                  title="Scatter Plot of Residual Sugar vs. Quality")
thinkplot.Show()

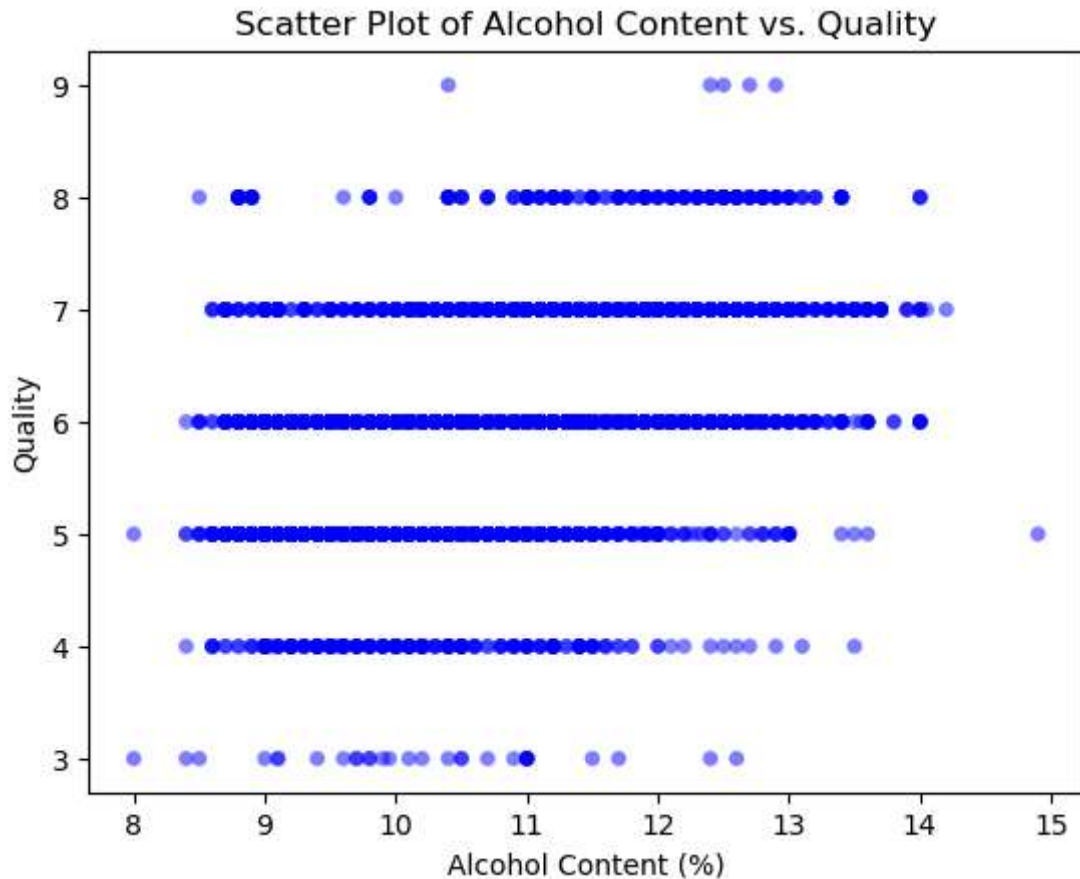
# Calculate Covariance and Pearson's Correlation for both pairs
# Pair 1: Alcohol Content vs. Quality
cov_alcohol_quality = np.cov(alcohol, quality)[0, 1]
pearson_alcohol_quality = stats.pearsonr(alcohol, quality)[0]

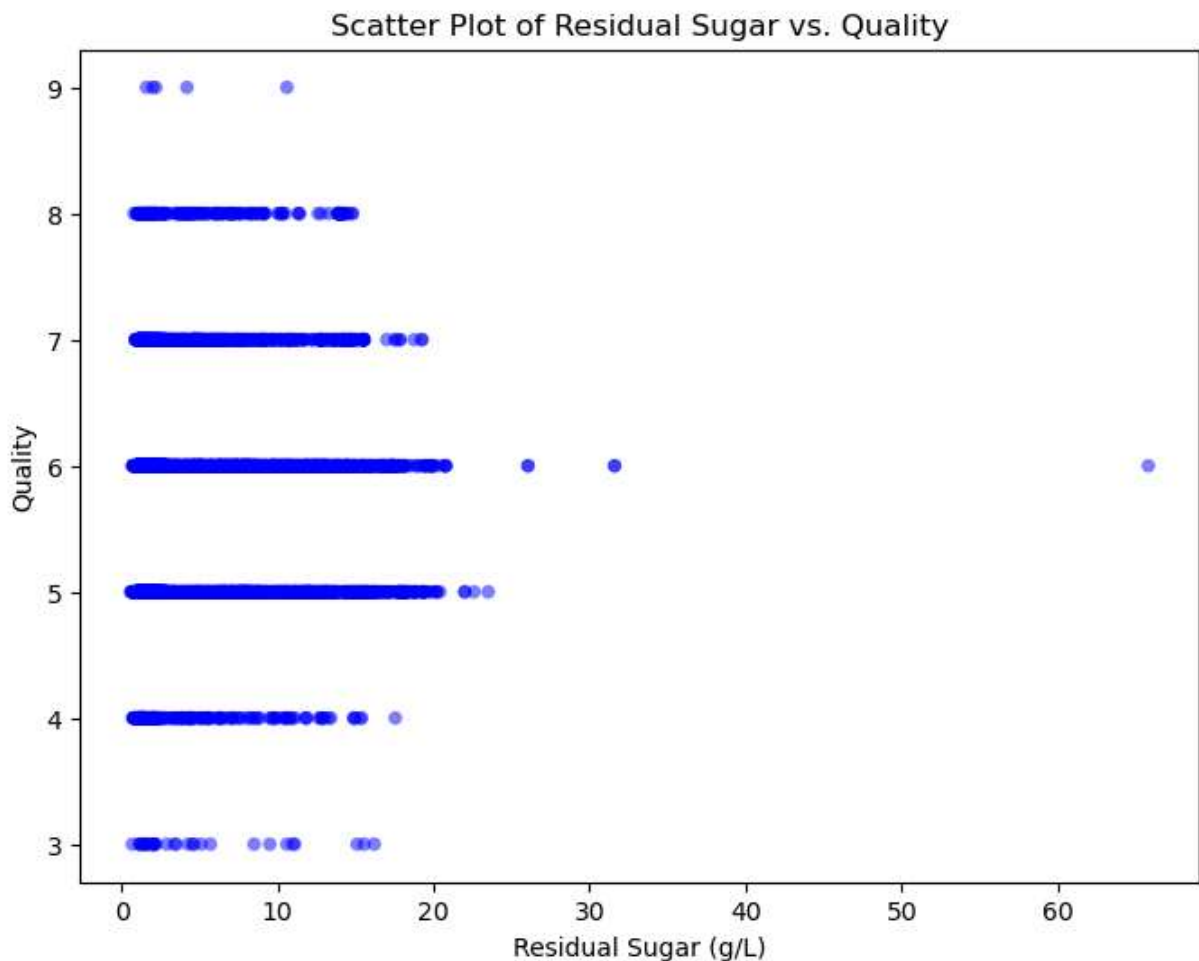
# Pair 2: Residual Sugar vs. Quality
cov_sugar_quality = np.cov(residual_sugar, quality)[0, 1]
pearson_sugar_quality = stats.pearsonr(residual_sugar, quality)[0]

print("Alcohol vs. Quality")
print(f"Covariance: {cov_alcohol_quality:.2f}")
print(f"Pearson's Correlation: {pearson_alcohol_quality:.2f}")

print("\nResidual Sugar vs. Quality")
print(f"Covariance: {cov_sugar_quality:.2f}")
print(f"Pearson's Correlation: {pearson_sugar_quality:.2f}")

```





Alcohol vs. Quality
 Covariance: 0.46
 Pearson's Correlation: 0.44

Residual Sugar vs. Quality
 Covariance: -0.15
 Pearson's Correlation: -0.04
 <Figure size 800x600 with 0 Axes>

Alcohol vs. Quality: The scatter plot points are fairly dispersed but has a slight upward trend toward the right side, indicating that wines with higher alcohol content may be associated with higher quality scores. However, it is not a strong relationship. The positive covariance of 0.46 means that as alcohol content increases, quality does too. However, this alone doesn't tell us the strength or consistency of the relationship. Pearson's correlation of 0.44 suggests a moderate positive, linear relationship between alcohol content and quality. It is still not very strong and implies there may be other factors that influence quality.

Residual Sugar vs. Quality: The scatter plot does not show us a clear trend and the scores are spread out across residual sugar levels. The negative covariance is very low, indicating a very minimal association between residual sugar and quality. Furthermore, Pearson's Correlation of -0.04 is close to 0, suggesting no linear relationship between these variables.

```
In [143... # Define the threshold for high and low alcohol (e.g., median split)
median_alcohol = complete_df['alcohol'].median()
```

```

high_alcohol_group = complete_df[complete_df['alcohol'] >= median_alcohol]['quality']
low_alcohol_group = complete_df[complete_df['alcohol'] < median_alcohol]['quality']

class DiffMeansPermute(thinkstats2.HypothesisTest):

    def TestStatistic(self, data):
        group1, group2 = data
        test_stat = abs(group1.mean() - group2.mean())
        return test_stat

    def MakeModel(self):
        group1, group2 = self.data
        self.n, self.m = len(group1), len(group2)
        self.pool = np.hstack((group1, group2))

    def RunModel(self):
        np.random.shuffle(self.pool)
        data = self.pool[:self.n], self.pool[self.n:]
        return data

# Use DiffMeansPermute to perform the permutation test and get the p-value
data = (high_alcohol_group, low_alcohol_group)
ht = DiffMeansPermute(data)
p_value = ht.PValue(iters=1000) # Run 1000 permutations

# Print the observed difference and p-value
print(f"Observed Difference in Mean Quality: {observed_diff:.2f}")
print(f"p-value: {p_value:.4f}")

```

Observed Difference in Mean Quality: 0.67
p-value: 0.0000

Wines with higher alcohol content have a quality rating that is 0.67 points higher than wines with lower alcohol content. The p-value of 0.0000 is suggesting a very strong statistical significance. There is a very low probability that the observed difference of 0.67 is due to random chance.

In [151...

```

# Start with a single-variable regression of quality vs alcohol content

import statsmodels.formula.api as smf

# Single-variable linear regression with Quality as the dependent variable and Alco
model_single = smf.ols(formula='quality ~ alcohol', data=complete_df).fit()

# Print the summary of the single-variable regression model
print(model_single.summary())

```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          quality    R-squared:                0.197
Model:                  OLS        Adj. R-squared:            0.197
Method:                 Least Squares    F-statistic:             1598.
Date:                   Mon, 11 Nov 2024    Prob (F-statistic):      1.50e-312
Time:                   19:53:17          Log-Likelihood:          -7623.4
No. Observations:       6497            AIC:                    1.525e+04
Df Residuals:           6495            BIC:                    1.526e+04
Df Model:               1
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.4053	0.086	27.988	0.000	2.237	2.574
alcohol	0.3253	0.008	39.970	0.000	0.309	0.341

```

=====
Omnibus:                123.922    Durbin-Watson:           1.636
Prob(Omnibus):           0.000    Jarque-Bera (JB):        258.800
Skew:                    0.031    Prob(JB):                6.34e-57
Kurtosis:                3.976    Cond. No.                 94.3
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

R-squared of 0.197 tells us that 19.7% of the variability in wine quality can be explained by the alcohol content. This indicates that alcohol content is a strong predictor of quality.

Furthermore, the coefficient for alcohol tells us that for every 1% increase in alcohol content, we can expect the quality rating to increase by 0.33 points. This relationship is statistically significant with a p-value of less than 0.05.

```

In [173... # Multiple-Variable Regression of Quality vs other variables

# Including Alcohol, Residual Sugar, pH, and Sulphates as explanatory variables
model_multiple = smf.ols(formula='quality ~ alcohol + residual_sugar + pH + fixed_a

# Print the summary of the multiple-variable regression model
print(model_multiple.summary())

```

OLS Regression Results

Dep. Variable:	quality	R-squared:	0.221			
Model:	OLS	Adj. R-squared:	0.221			
Method:	Least Squares	F-statistic:	368.7			
Date:	Tue, 12 Nov 2024	Prob (F-statistic):	0.00			
Time:	16:38:34	Log-Likelihood:	-7525.8			
No. Observations:	6497	AIC:	1.506e+04			
Df Residuals:	6491	BIC:	1.510e+04			
Df Model:	5					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	2.3639	0.261	9.073	0.000	1.853	2.875
alcohol	0.3628	0.009	41.728	0.000	0.346	0.380
residual_sugar	0.0264	0.002	11.523	0.000	0.022	0.031
pH	-0.1668	0.067	-2.495	0.013	-0.298	-0.036
fixed_acidity	-0.0317	0.008	-3.795	0.000	-0.048	-0.015
sulphates	0.5089	0.071	7.202	0.000	0.370	0.647
=====						
Omnibus:	124.457	Durbin-Watson:	1.653			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	260.837			
Skew:	0.028	Prob(JB):	2.29e-57			
Kurtosis:	3.980	Cond. No.	402.			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The R-squared of 0.221 tells us that 22.1% of the variability in wine quality can be explained by the combination of alcohol content, fixed acidity, residual sugar, pH, and sulphates. This is a moderate level of explanatory power, suggesting that other factors could also contribute to wine quality. The coefficient for alcohol is 0.3628, which tells us that for every 1% increase in alcohol content, the quality increases by about 0.36 points. The next significant variable is sulphates, with a coefficient of 0.5089, and a p-value that indicates this is statistically significant.

Alcohol and sulphates are the strongest predictors of wine quality based on our multiple regression model. Residual sugar shows a small impact, and pH does not show a statistically significant relationship.